Material Transfer Robot

ROS2 Topics & Services API Reference

This document lists all ROS2 topics and services available for UI integration with the robot delivery system. Use this as the interface contract between the robot stack and the frontend dashboard.

Topics

All topics published by the robot stack. The UI subscribes to these for live status updates.

Topic	Message Type	Publisher	Rate	Data Fields
`/job_status`	`job_manager/JobStatus`	job_manager	On change	header, job_id, pickup_room, dropoff_room, priority, state (uint8), state_name (string), message
`/system/active_jobs`	`std_msgs/Int32`	job_manager	1 Hz	data (count of queued + active jobs)
`/job/awaiting_confirmation`	`std_msgs/String`	job_manager	On event	data (stage, location, timeout info). Published when robot arrives at pickup or dropoff.
`/robot_health`	`system_supervisor/RobotHealth`	supervisor	1 Hz	header, cpu_percent, cpu_count, memory_used_mb, memory_total_mb, nodes[] (per-node health), system_state, supervisor_state, system_message, autonomous_enabled
`/system/ready`	`std_msgs/Empty`	supervisor	Once	Published when all always-on nodes are up. Robot is ready to accept jobs.

Services

Services the UI can call to interact with the robot. All services are synchronous request/response.

Service	Type	Server	Request / Response
`/request_delivery`	`job_manager/DeliveryJob`	job_manager	Req: pickup_room (string), dropoff_room (string), priority (uint8: 0=normal, 1=high) Resp: job_id (string), accepted (bool), message (string)
`/cancel_job`	`job_manager/CancelJob`	job_manager	Req: job_id (string, empty = cancel current) Resp: success (bool), message (string)
`/job/confirm`	`job_manager/ConfirmJob`	job_manager	Req: proceed (bool, true=continue, false=abort) Resp: success (bool), message (string)
`/navigate_to_room`	`mission_controller/NavigateToRoom`	mission_controller	Req: room_name (string) Resp: success (bool), message (string)

UI Workflow

Submitting a Delivery

1. Call /request_delivery with pickup_room, dropoff_room, and priority.
2. Service returns job_id and accepted status immediately.
3. Subscribe to /job_status to track the job through its lifecycle.
4. When state changes to PICKUP_ARRIVED, the /job/awaiting_confirmation topic fires.
5. Show a 'Confirm Pickup' button. When clicked, call /job/confirm with proceed: true.
6. Robot proceeds to dropoff. Same confirmation flow at DROPOFF_ARRIVED.
7. Job completes. If no more jobs, robot returns home automatically.

Cancelling a Job

Call /cancel_job with the job_id. If job_id is empty, cancels the currently active job. Queued jobs are removed immediately. Active jobs stop after the current navigation step.

Monitoring System Health

Subscribe to /robot_health (1 Hz) for system-wide status. The system_message field gives a human-readable summary. The supervisor_state field indicates the current lifecycle phase.

Subscribe to /system/ready (published once) to know when the robot is fully booted and ready to accept jobs.

Job State Reference

States published in the /job_status topic's state field:

Value	State Name	Description
0	`QUEUED`	Job accepted, waiting in queue
1	`WAITING_FOR_NAV`	Waiting for supervisor to bring up navigation stack
2	`PICKUP_NAV`	Navigating to pickup room
3	`PICKUP_ARRIVED`	At pickup location — awaiting confirmation via /job/confirm
4	`DROPOFF_NAV`	Navigating to dropoff room
5	`DROPOFF_ARRIVED`	At dropoff location — awaiting confirmation via /job/confirm
6	`RETURNING`	Returning to home position
7	`COMPLETE`	Delivery finished successfully
8	`FAILED`	Delivery failed (navigation error, timeout, etc.)
9	`CANCELLED`	Job cancelled by user or pickup timeout

Supervisor State Reference

States published in the /robot_health topic's supervisor_state field:

Value	State	Description
0	`BOOTING`	Waiting for all always-on nodes to start (motors, teleop, encoders, etc.)
1	`IDLE`	All always-on nodes running. Waiting for delivery jobs.
2	`ACTIVATING`	Starting on-demand nodes: ZED → odom → Nav2 → PID → mission
3	`ACTIVE`	Full navigation stack running. Executing delivery jobs.
4	`DEACTIVATING`	Shutting down on-demand nodes. Returning to IDLE.

Confirmation Behavior

At Pickup

Robot waits for /job/confirm call. If proceed: true, continues to dropoff. If proceed: false or 5-minute timeout with no response, the job is CANCELLED.

At Dropoff

Robot waits for /job/confirm call. If proceed: true, marks job COMPLETE. If proceed: false or 5-minute timeout, marks job COMPLETE anyway (item left at dropoff).

Quick Test Commands

Submit a delivery:

```
ros2 service call /request_delivery job_manager/srv/DeliveryJob "{pickup_room: 'H201',
dropoff_room: 'H207', priority: 0}"
```

Confirm pickup/dropoff:

```
ros2 service call /job/confirm job_manager/srv/ConfirmJob "{proceed: true}"
```

Cancel current job:

```
ros2 service call /cancel_job job_manager/srv/CancelJob "{job_id: ''}"
```

Watch job status:

```
ros2 topic echo /job_status
```

Watch system health:

```
ros2 topic echo /robot_health
```